

SAI Happy Farms Farm ERP — Supervisor User Manual

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Target Audience: Operational Farm Supervisors

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1. Introduction

As a Supervisor for SAI Happy Farms, your primary responsibility is to monitor daily operational metrics for the farms explicitly assigned to you. This ERP system centralizes real-time data submitted by farmers, allowing you to instantly track mortality, feed stock, and egg production without waiting for weekly paper reports.

2. Supervisor Login

4. Navigate to the application URL on your desktop, tablet, or mobile device.
5. Enter your secure Email and Password (passwords must be ≥ 8 characters).
6. Upon login, the system will securely verify your role and map you only to your assigned farms.

3. Enterprise Analytics Dashboard

Your main workspace is the Enterprise Analytics Dashboard. It is designed to provide immediate visual insights based on real-time data.

Dashboard Components:

- Summary KPI Cards: Quick numeric readouts for current master stock and production.
- Analytics Charts: 8 visual Recharts mapping trends such as Production %, Mortality, and Feed.
- Farm Performance Comparison: A ranking table scoring your farmers based on their performance.
- Alert Feed: Inline alerts highlighting critical operational issues.

4. Assigned Farms & Filtering

The dashboard strictly enforces data privacy. You can only view data for farms mapped to your account.

- Farm Filter: Use the top dropdown to view data for a specific farm or select "All Assigned Farms" to view aggregated data.
- Date Filter: Switch between 7-Day and 30-Day historical views. Selecting a timeframe will instantly query the database and re-render all visual charts.

5. KPIs & Analytics

The system automatically computes Key Performance Indicators (KPIs) in real-time utilizing the daily logs submitted by farmers.

Calculated Formulas

- Mortality %: $\text{(Daily Mortality / Current Bird Count)} * 100$
- Culling %: $\text{(Daily Culling / Current Bird Count)} * 100$
- Egg Production %: $\text{(Eggs Produced / Current Bird Count)} * 100$
- Selection Egg %: $\text{(Selection Eggs / Total Eggs Produced)} * 100$
- Feed per Bird: $\text{(Feed Kg * 1000) / Current Bird Count}$ (Output in grams)
- Feed per Egg: $\text{Total Feed Kg / (Total Eggs / 12)}$ (Output in Kg/Dozen)

These metrics allow you to track real-world health and output without manual mathematics.

6. Farm Performance Ranking

The dashboard includes a Farm Performance Comparison table. This engine ranks your assigned farms based on a calculated Farm Performance Score.

The score evaluates:

- Production Gap (How close actual production is to standard curves)
- FCR (Feed Conversion Ratio)
- Egg Damage/Selection %
- Mortality %

(Note: Specific algorithm weightings are driven internally by the analytics engine. Detailed tie-break conditions are not verified from the available project evidence).

7. Alerts & Monitoring

The dashboard will actively flag conditions requiring your attention. These in-app visual alert cards will appear when:

- Missing Submissions: A farmer has failed to submit their daily report.
- Mortality Spikes: The daily mortality count indicates a potential health crisis.
- Production Drops: Egg production suddenly falls below acceptable thresholds.

Note: The alerts are visible only while you are actively viewing the dashboard. The system does not currently send external SMS or Email notifications (Not Implemented).

8. Supervisor Restrictions

To maintain system integrity, your Supervisor account is subject to strict security constraints:

What you CAN do:

- View dashboards and charts for your assigned farms.
- Monitor live KPIs and farm performance rankings.
- Filter historical data by 7-day and 30-day windows.

What you CANNOT do:

- You cannot edit or delete farmer data submissions.
- You cannot view data for farms that belong to another supervisor. The database will explicitly block the connection (`HTTP 403`).
- You cannot add new farmers, reassign farms, or execute master data imports. Those functions require Administrator privileges.

9. Supervisor Testing Checklist

Test ID	Module	Steps	Expected Result	Pass/Fail
SUP-001	Farm Access	Log in and open Farm filter dropdown.	Only assigned farms are listed.	
SUP-002	Date Filter	Switch filter from 7-day to 30-day view.	Charts instantly redraw for the 30-day period.	
SUP-003	Authorization	Attempt to directly view a URL/ID of an unassigned farm.	Blocked by Firestore security rules.	
SUP-004	Alert Visibility	Locate a farm with a missed submission.	"Missing Submission" alert card rendered.	

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